A Geometric Algebra-Based Model for Enhanced Hyperspectral Anomaly Detection

Yilin Liu, Yong Luo, Rui Wang, *Senior Member, IEEE*, Yao Huang, Ming Ju, and Wei Xiang, *Senior Member, IEEE*

***Abstract*— Hyperspectral anomaly detection (HAD) is of great significance in remote sensing by identifying spectrally distinct pixels as anomalies without prior information, while the majority of pixels with similar spectral characteristics are classified as background. While existing approaches combine the strengths of deep learning (DL) in feature extraction and background suppression with the effectiveness of low-rank representation (LRR) in background modeling, they fail to fully exploit rich spatial information and long-range spectral dependencies due to the limitations of conventional convolutional operations. To address this issue, we propose a geometric algebra (GA)-based deep neural network for HAD, named GA-HAD. The network constructs an encoder–decoder architecture using GA convolutional layers to extract spectral–spatial features, capturing multidimensional spatial information while preserving the intricate spectral characteristics of hyperspectral images (HSIs). A specialized reconstruction error is designed to train the GA feature extraction network with high efficiency and accuracy. The encoder features are integrated with the LRR algorithm and a Gaussian mixture model (GMM)-based dictionary to improve background modeling and robustness. Finally, the detection maps output by the low-rank detection module are refined through an edge-preserving filter to produce the final detection result. Extensive experiments on four datasets demonstrate the superiority of our proposed GA-HAD framework in overall detection effect compared to thirteen representative baselines, implying our potential application in the field of HAD.**

***Index Terms*— Anomaly detection, geometric algebra (GA), hyperspectral image (HSI), spectral–spatial feature extraction.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spatial and spectral information of various land covers, which enable the detection of subtle differences between different objects and scenes. Depending on the availability of prior spectral information, HSI detection can be categorized into supervised target detection and unsupervised anomaly detection [1]. The former uses known spectral signatures, while the latter classifies distinct pixels without prior information. As an unsupervised technique, hyperspectral anomaly detection (HAD) aims to detect pixels with spectral signatures significantly different from the surrounding background, without relying on prior scene or environmental information. Pixels with similar spectra are classified as background. HAD has been widely applied in vegetation monitoring [2], geological exploration [3], and military reconnaissance [4].

In the field of HAD, the main objective is to identify pixels that have sparsity and significantly different characteristics from the background and to classify them accordingly as anomalies. In recent decades, a multitude of algorithms have been proposed to enhance the performance of HAD, focused on efficiently handling high-dimensional data, accurately detecting anomalies, effectively suppressing background, and enhancing the distinctiveness between background and anomalies. Initially, statistical algorithms such as the Reed–Xiaoli (RX) algorithm [5] were primarily leveraged for HAD, which models the background pixels as a specific distribution and identifies anomalies as pixels that fall out of the background distribution. Representation-based methods later improved this by constructing background dictionaries from representative background pixels and classifying anomalies as those poorly represented by the dictionary. In recent years, deep learning (DL) algorithms have gained extensive application in the field of HAD due to their strong feature extraction abilities. DL-based approaches include using an autoencoder (AE) to enhance background–anomaly differentiation and leveraging generative adversarial networks (GANs) to more accurately model background distributions.

A. Motivation

DL and low-rank representation (LRR) are two approaches to HAD that have been widely studied and adopted in recent years. DL-based methods are leveraged to suppress background through automatic feature extraction and complex nonlinear transformations but tend to suppress anomaly highlighting due to complex backgrounds and sparse anomaly data. Conversely, LRR can describe the background distribution via a background dictionary, effectively handling complex hyperspectral data. However, directly decomposing redundant original data can compromise the separability of background and anomalies.

To leverage the strengths of DL and LRR methods, researchers [6] have combined them for anomaly detection. Specifically, the latent feature matrix extracted by the DL network replaces the original hyperspectral data as input to LRR. Despite the improvement that enhances the ability to handle

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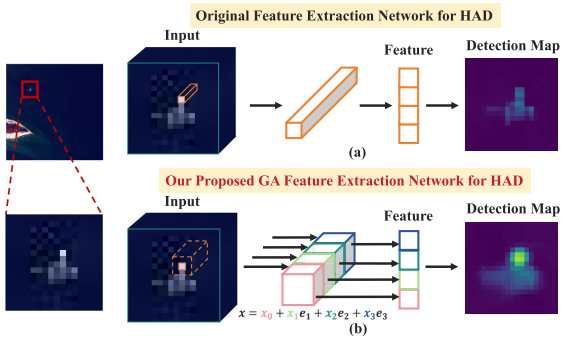


Fig. 1. Comparison of the previous original feature extraction network and our proposed network. (a) Original feature extraction network. (b) Our GA feature extraction network.

complex backgrounds, the integrated approach still struggles to effectively emphasize anomalies. Through our analysis, we found that the problem is due to the fixed size of the convolution kernel and that the 1-D convolution kernel can only extract short-range spectral information in the spectral dimension. As shown in Fig. 1(a), HSIs usually contain hundreds of spectral channels, making ordinary convolutional layers challenging to capture long-range dependencies between spectral dimensions. Additionally, most studies have used only 1-D spectral convolution for feature extraction, ignoring the rich spatial information of HSIs. However, the simplistic utilization of spatial information, which involves extensive neighborhood selection, simplistic averaging of spectral features, and inappropriate pixel weighting, can undermine the accuracy of anomaly detection by deteriorating anomaly discrimination.

B. Our Solution

To adapt to the numerous spectral channels in HSIs, we introduce geometric algebra (GA) theory [7]. As illustrated in Fig. 1(b), GA assigns multiple spectral channels of the HSI to the components of a GA multivector, effectively preserving structural information between spectral dimensions. By using GA multivectors and geometric product calculations, the network more effectively captures nonlinear relationships between spectral dimensions, preserving local and global structural information among spectral dimensions, and significantly enhancing the separability between background and anomalies. To fully leverage the rich spatial information of HSIs, we chose the GA-based convolutional neural network (GA-CNN) for feature extraction from HSIs. Existing research has shown that GA-CNN layers have achieved excellent results in hyperspectral image (HSI) processing, such as HSI classification [8].

We have specifically designed a GA-based neural network encoder–decoder to efficiently retain spectral structural information and effectively handle complex HSI scenarios, tailored to the characteristics of the HAD task. By feeding HSI cubes with spatial information into the GA encoder and reconstructing only the target pixel's spectral information, we maintain feature distinctiveness while incorporating spatial context. In addition, we designed a dedicated reconstruction error for training the GA feature extraction network to improve training efficiency and accuracy. The encoder features are integrated with the LRR algorithm, enhanced by a Gaussian mixture model (GMM)-based dictionary to improve background modeling and robustness. Inspired by LREN [6], we train feature extraction and dictionary construction together, addressing the challenge that the extracted features may not be the optimal solution for the inference process model. Finally, edge-preserving filtering refines the detection results by using spatial correlations.

C. Contributions

Our main contributions to this work are summarized as follows.

1) We propose an unsupervised and trainable network for HAD. This network integrates feature learning and anomaly detection with the same framework, delivering enhanced detection accuracy.
2) A GA-based codec network is developed to exploit spatial information in HSIs, representing the first application of GA theory in the field of HAD. This network is designed to extract spectral–spatial features for anomaly detection, improve training efficiency through a unique reconstruction error, and better preserve inherent spectral information during training.
3) The anomaly detection inference method that utilizes LRR is integrated with the built GA-based codec network. This approach combines the strengths of LRR and DL, improving the distinction between background and anomalies and using the advantages of GA to highlight anomalies more clearly.
4) To validate the effectiveness of the proposed method, it was compared with other HAD algorithms on typical datasets. The results demonstrate the superior performance of the proposed method.

II. Related Work

Various HAD algorithms have been proposed and can be broadly divided into three main categories [9], namely statistics-based methods, representation-based methods, and DL-based methods.

A. Statistics-Based Methods

The statistics-based HAD method assumes that background pixels in HSIs follow a specific statistical model and then marks pixels that do not conform to the model as anomalies. The RX algorithm [5], commonly categorized into global RX (GRX) and local RX (LRX) [10], is a representative statistics-based algorithm. The RX algorithm models the background as a multivariate Gaussian distribution and detects anomalies by calculating the Mahalanobis distance between test samples and the background distribution. Due to the complex spectral–spatial structure of HSIs, the assumption that the background follows a Gaussian distribution rarely holds [11]. To address these challenges, researchers have proposed a series of enhancements based on the RX algorithm, such as kernel-RX (KRX) [12], weighted RX (WRX), and linear filter-based RX (LF-RX) [13]. LSAD [14] further improves LRX by incorporating a multiwindow strategy and subspace projection, while the robust nonlinear anomaly detection (RNAD)

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method [15] addresses the issue of anomaly contamination with statistical modeling. To address the challenge of detecting multipixel anomalies, researchers have recently introduced a two-step GLRT (2S-GLRT) [16] method. The kernel isolation forest detection (KIFD) [17] assumes that anomalies are more easily distinguishable from the background in kernel space and employs an isolation forest to identify these isolated samples. In complex backgrounds, statistics-based methods struggle to model the background accurately, limiting detection performance.

B. Representation-Based Methods

Representation-based methods avoid assuming a statistical background distribution in HSIs by representing each pixel as a linear combination of background dictionary atoms, with anomalies detected based on reconstruction residuals. These methods include sparse representation (SR), cooperative representation (CR), and LRR, differing in regularization constraints. The SR methods, initially proposed by Chen et al. [18] for hyperspectral target detection, were later adapted to HAD by Li et al. [19]. SSEAD [20] evaluates the anomaly scores by learning the frequencies of dictionary atoms through SR and incorporating enhancement strategies. Li and Du [21] integrated CR with spatial information, detecting anomalies based on the reconstruction error of neighboring pixels. Lin and Lin [22] proposed SuperRPCA by integrating the collaborative representation detector (CRD) into the robust principal component analysis (RPCA) [23], utilizing local pixel information to significantly reduce computational costs. The LRR methods, initially proposed by Liu et al. [24] for subspace clustering, have been applied to HAD and further refined by various scholars. The low-rank and SR (LRASR) [25] method detects anomalies by exploiting the low-rank property of background dictionaries and the sparsity of anomalies. To improve background suppression, the effective anomaly space (EAS) [26] using independent component analysis and sparsity cardinality has been proposed to improve LRASR-based methods. The low-rank and sparse matrix decomposition (LRaSMD) [27] method separates anomalies by decomposing hyperspectral data into low-rank, sparse, and noise components through convex optimization. Building on this foundation, the LSMAD [11] method uses the classic GoDec algorithm [28] to solve for the low-rank and sparse components and applies Mahalanobis distance differences to identify anomalies. The antinoise hierarchical mutual-incoherence-induced discriminative learning (AHMID) [29] method is proposed to enhance the separability between the background and anomalies. Several approaches have enhanced model accuracy by integrating more appropriate constraints into the LRR model [30], [31]. GSEAD [32] utilizes graph-based modeling and manifold learning to capture intrinsic data structures, thereby enhancing its adaptability to diverse HSI conditions. Some methods focus on improving background estimation. RBRSE [33] constructs a kernel dictionary through robust background regression, while SVBS [34] enhances background estimation by isolating slowly varying signals. Representation-based methods simplify background modeling but struggle with complex nonlinear features.

C. DL-Based Methods

DL algorithms have been widely applied in HAD in recent years. AE [35] trains anomaly detection networks using pixel reconstruction errors. The robust graph AE (RGAE) [36] preserves the geometric spatial structure of HSIs by embedding a graph regularization term based on superpixel segmentation. Auto-AD [37] reconstructs the background using a fully convolutional AE with skip connections. The deep feature aggregation network (DFAN) [38] employs orthogonal spectral attention and background pattern modeling, whereas FCAE-DCAC [39] integrates spatial–spectral attention with dual-clustering and adversarial consistency for improved AE-based anomaly detection. The convolutional neural network (CNN)-based HAD method [40] trains models with similar and dissimilar pixel pairs and evaluates anomalies based on average similarity to surrounding pixels. The spectral–spatial deep support vector data description (ssDSVDD) algorithm [41] uses a dual-stream deep convolutional AE network to extract spectral–spatial features from HSIs. Song et al. [42] integrated the CNN with LRR to improve the detection accuracy. The low-rank embedding network (LREN) [6] jointly trains the AE and GMM to enhance model adaptability. The HADGAN [43] employs an AE to enforce significant reconstruction constraints on the GAN. Moreover, a dual-GAN-based algorithm [44] is proposed, demonstrating the capability to more authentically reconstruct the background distribution. Other DL methods have been proposed for use with HAD. The GT-HAD method [45] uses spatial–spectral similarity in HSIs to guide background and anomaly reconstruction through a Transformer network equipped with gating units. Gao et al. [46] applied a blind-spot self-supervised learning network to HAD, predicting the central pixel using surrounding pixels [47], with later improvements enhancing detection for large anomalies [48], [49], [50].

Many methods treat HSIs as third-order tensors and use tensor decomposition algorithms for anomaly detection, such as those based on Tucker decomposition [51], [52], [53]. The prior-based tensor approximation (PTA) [54] significantly improves the ability to detect anomalies of different sizes by applying different regularization constraints to the background and anomaly tensors. PCA-TLRSR [55] is a tensor low-rank sparse representation method that uses principal component analysis (PCA) as a preprocessing step and uses the tensor nuclear norm (TNN) to overcome the problem of predefined ranks inherent in Tucker decomposition.

III. Preliminaries

A. GA Background

GA [56], also known as Clifford Algebra, was introduced by Clifford in the late 19th century. It integrates the concepts of quaternions proposed by Hamilton and exterior algebra developed by Grassmann [57].

GA provides an efficient framework for geometric calculations without dependence on coordinate information. In this framework, all geometric objects are the most fundamental entities in algebra. GA facilitates the computation and analysis

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of multidimensional signals, reducing computational complexity and enhancing efficiency. The conventional definition of GA is carried out in the context of vector spaces endowed with an inner product [7].

Mathematically, suppose that $\mathbb{G}_{p,q,r}$ denotes the unique GA set for an n-dimensional quadratic vector space $\mathbb{R}^{p,q,r}$, where $n = p + q + r$, and $p, q, r \in \{0, \ldots, n\}$. Here, p and q represent the counts of vectors with positive and negative squares in the n-dimensional space, respectively. The particular selection of algebra, determined by p, q, and r, enables the effective modeling of many types of geometry. For all $i \in \{1, \ldots, n\}$, the following condition is fulfilled:

$$e_i^2 = \begin{cases} 1, & 1 \le i \le p \\ -1, & p+1 \le i \le p+q \\ 0, & p+q+1 \le i \le p+q+r = n. \end{cases} \tag{1}$$

Due to the focus on the case of $q = 0$ and $r = 0$ in this article, we use $\mathbb{G}_n$ to represent $\mathbb{G}_{p,0,0}$ and $\mathbb{R}^n$ to represent $\mathbb{R}^{p,0,0}$. Denote by $\mathbf{b}$ the set of all naturally ordered subsets of the power set $P(\{1, \ldots, n\})$

$$\mathbf{b} := \{\{i_1, \ldots, i_s\} \in P(\{1, \ldots, n\}) | 1 \le i_1 \le \cdots \le i_s \le n\}. \tag{2}$$

Furthermore, let $\{e_1, e_2, \ldots, e_n\}$ be an orthogonal basis vector in the GA space $\mathbb{R}^n$. Now define, for all $I \in \mathbf{b}$, e_I to be the geometric product

$$e_I := \left\{e_{i_1}, \ldots, e_{i_s} | I \in \mathbf{b}\right\} \tag{3}$$

where $e_\emptyset$ is typically set to 1 [58].

Multivectors are fundamental elements in the GA space. For any two multivectors x and y in space $\mathbb{G}_n$, their geometric product can be defined as

$$x \otimes_{p,q} y = x \cdot y + x \wedge y \tag{4}$$

where $x \cdot y$ represents the inner product of two vectors and $x \wedge y$ represents the outer product of two vectors.

In this article, since all basis vectors are orthogonal, $e_i e_j = e_i \cdot e_j + e_i \wedge e_j = e_i \wedge e_j$, making the geometric product equivalent to the outer product. Therefore, the geometric product is also noncommutative

$$e_i e_j = e_{ij} = -e_j e_i = -e_{ji}, \quad i, j = 1, \ldots, n, i \ne j. \tag{5}$$

In space $\mathbb{R}^n$, there exists complete orthogonal basis blades of size 2^n. Taking space $\mathbb{R}^3$ as an example, the complete orthogonal basis of this space is composed of $2^3 = 8$ vectors, denoted by $\{1, \{e_1, e_2, e_3\}, \{e_1e_2, e_1e_3, e_2e_3\}, \{e_1e_2e_3\}\}$, and its simplified form can be expressed as $\{1, e_1, e_2, e_3, e_{12}, e_{13}, e_{23}, e_{123}\}$.

A product of k basis vectors forms a basis k-blade, where the grade of the blade corresponds to the dimension of the subspace it represents. In this context, vectors composed of basis e_i components are one-blades. Meanwhile, two-blades are formed in the pattern $e_i e_j$, and the sequence continues for higher dimensions. We refer to k-vectors as a homogeneous combination of basis blades of grade k [59].

The fundamental elements in GA are multivectors, typically expressed in terms of a complete orthogonal basis in the GA space. A multivector $x \in \mathbb{G}_{p,q,r}$ is a sum of k-vectors. Specifically, any multivector in GA can be written as

$$x = \sum_{k=0}^{n} \langle x \rangle_k = \sum_{I \in \mathbf{b}} [x]_I e_I \tag{6}$$

where $\langle x \rangle_k$ denotes the k-vector part of x and $[x]_I \in \mathbb{R}$ represents the coefficients of the orthogonal basis vectors.

For instance, the fundamental form of a multivector in $\mathbb{G}_3$ can be written as

$$\begin{aligned} x &= \langle x \rangle_0 + \langle x \rangle_1 + \langle x \rangle_2 + \langle x \rangle_3 \\ &= x_0 1 + x_1 e_1 + x_2 e_2 + x_3 e_3 \\ &\quad + x_{12} e_{12} + x_{13} e_{13} + x_{23} e_{23} + x_{123}. \end{aligned} \tag{7}$$

The GA space consists of $n + 1$ subspaces. Taking $\mathbb{G}_3$ as an example, $x_0 1$ is in scalar form, an element of the 0-grade subspace. $x_1 e_1 + x_2 e_2 + x_3 e_3$ is a linear combination of one-blades, which is referred to as a vector. $x_{12} e_{12} + x_{13} e_{13} + x_{23} e_{23}$ is called a bivector, an element of the two-grade subspace. Its geometric interpretation is a vector plane spanned by two vectors. $x_{123} e_{123}$, in the form of three blades, is dubbed a trivector, with a geometric interpretation of a volume element spanned by a vector plane and a 1-D vector.

B. GA Neural Networks

A GA neural network is the outcome of extending the CNN from the real number domain to the GA domain, where all real-valued multiplications are replaced by geometric products. A real linear neuron can be regarded as the initial instance of a GA neuron. A GA neuron performs the following computations mapping from $(\mathbb{G}_{p,q,r})^n$ to $\mathbb{G}_{p,q,r}$

$$y = \sum_{i=1}^{n} \omega_i \otimes_{p,q,r} x_i + b \tag{8}$$

where x_i, ω_i, and $b \in \mathbb{G}_i$ denote the inputs, weights, and bias, respectively.

Based on GA neurons, researchers have also proposed an operational model for GA linear and convolutional layers, which are analogous to ordinary linear and convolutional layers in the real-number domain. This operational model is schematically shown in Fig. 2. The GA linear layer takes the multivector of a single pixel as input for extracting the spectral features. In contrast, the GA convolutional layer accepts as input a multivector of cubes of pixels, which can be extracted to spectral–spatial features. The GA convolutional layer can extract information from multichannel signals. In the operations of the GA convolutional layer, all elements are in the form of multivectors, and all multiplication operations are replaced by geometric product operations [60]. Similar to real-number domain CNNs, GA convolutional layers also require parameters of convolutional kernel size, stride, and padding.

For instance, consider an input HSI cube of size [3, 3, 120]. In a traditional real-valued convolutional layer using 64 3 $\times$ 3 convolution kernels, the 120-D spectral data is directly processed, resulting in an output of size [1, 1, 64]. By contrast, for a GA convolutional layer with 16 3 $\times$ 3 convolution kernels, the actual structure is [3, 3, 16, 4]. This means that

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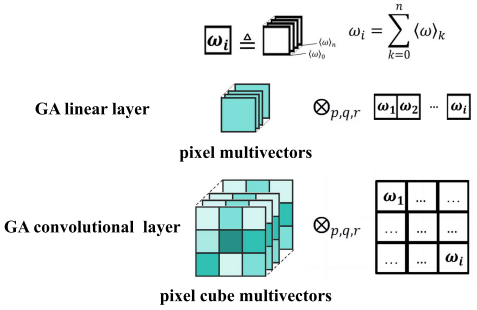


Fig. 2. Operational model for the GA linear layer and the GA convolutional layer.

each convolution kernel spatial dimension is of [3, 3], with a total of 16 output channels, and each channel data is divided into four components of a GA multivector. Specifically, the original 120-D spectral vector is evenly split into four parts, each of which is assigned to one component of the multivector, forming an input format of [3, 3, 30, 4]. After performing GA convolution operations, the output is a multivector of size [1, 1, 16, 4].

Based on these parameters, the multivectors of the pixels containing the complete receptive field in the input feature map, and the weight multivectors of the corresponding positions of the convolutional kernel, undergo geometric product operations. In GA convolutional layers, the operations in the spectral dimension depend on the number of convolutional kernels. The properties of GA enable efficient preservation of structural information between spectral bands during neural network processing. The use of GA convolutional layers can then combine the pixel spatial information for feature extraction while preserving the original spectral structure information for enhanced detection.

IV. Proposed GA-HAD Method

In this section, we propose the GA-HAD method for HAD. The flowchart of the proposed method in the training stage is illustrated in Fig. 3. We will briefly introduce each module of the proposed network and elaborate on the implementation details.

A. Feature Extraction Through the GA Codec

Due to limitations in hyperspectral imaging equipment, raw HSI pixels often contain mixed image elements. Extracting features from these raw pixels can help identify essential features, increase the contrast between background and anomalies, reduce data dimension, and improve computational efficiency. To fully exploit this rich spectral–spatial information in HAD studies, we propose a module that uses a GA encoder to perform feature extraction on raw HSIs. The model parameters are optimally trained by the error between the pixel features reconstructed by the GA decoder and the original pixel spectral features. The network of the proposed method module consists of five layers. The first three layers belong to the GA encoder, which consists of one 1-D spectral convolutional layer, one GA convolutional layer, and one fully connected layer. The last two layers belong to the GA decoder, which are two GA linear layers. The spectral features of the HSI are initially extracted using the 1-D spectral convolutional layer, which matches the spectral dimensions with the dimensions required for the GA convolutional layer. The GA convolutional layer synthesizes spatial information to extract the spectral–spatial features of the image. The fully connected layer expands the GA features into 1-D vectors and determines the final number of extracted features. The GA decoder utilizes the GA linear layers to reconstruct pixel features and train the model using the reconstruction error. The GA codec module is an HSI cube-based feature extraction module that integrates spectral and spatial information, resulting in better performance than individual pixel-based features.

To extract features containing spatial information, we extract $H \times W$ HSI cubes centered on each original pixel with a window size of $w \times w$ from the original HSI of size $[H, W, C]$. Specifically, the pixels at the edge of the image are zeroed out according to the window size before extracting the image blocks. The extracted HSI cubes are then fed to the GA encoder. Denote by $\mathbf{H} = [\mathbf{h}_i \in \mathbb{R}^{w\times w\times C}]_{i=1}^{N}$ the HSI cubes, where $N = H \times W$ is the number of raw HSI pixels and $\mathbf{h}_i$ denotes the ith HSI cube centered on the ith original pixel.

To reduce computational costs, $\mathbf{h}_i$ is passed through a 1-D spectral convolutional layer to extract spectral features and yield $\mathbf{h}_o \in \mathbb{R}^{w\times w\times f}$, where f denotes the number of output channels and $o = 1, \ldots, N$. This operation ensures that the number f of spectral dimensions is a multiple of the number of orthogonal basis vectors of the chosen GA space, matching the number of input spectral dimensions required by the GA convolutional layer.

Before feeding the processed HSI cube $\mathbf{h}_o$ into the GA convolutional layer, each pixel must be converted into a GA multivector. This operation maps each spectrum of the HSI cube onto each component of the GA multivector separately. Each pixel of $\mathbf{h}_o$ can be represented by a multivector x_{ik} in $\mathbb{G}_n$

$$x_{ik} = \sum_{0\le p\le n} [x_{ik}]_p e_p + \sum_{0\le p\le q\le n} [x_{ik}]_{pq} e_{pq} + \cdots + [x_{ik}]_{1,\ldots,n} e_{1,\ldots,n} \quad (9)$$

where $[x_{ik}]_p$, $[x_{ik}]_{pq}$, and $[x_{ik}]_{1,\ldots,n}$ are the spectral components of the pixel at positions k in $\mathbf{h}_o$.

After that, the GA multivectors are fed into the GA convolutional layer as inputs. For the input receptive field elements centered at k, the output through the GA convolutional layer can be calculated as follows:

$$y_{ik} = g\left(\sum_{j}^{m\times d\times d} \left(\omega_j \otimes x_{ij} + b_j\right)\right), \quad i = 1, \ldots, N \quad (10)$$

where m is the number of input channels, d denotes a prespecified GA convolutional kernel size, and $\omega_j, b_j \in \mathbb{G}_n$. The activation function of the GA convolutional layer, denoted by g, is selected as the $tanh$ function, which is similar to that of a real-valued CNN.

The GA convolutional layer output is flattened to form a 1-D vector, which is passed through a fully connected layer to obtain the spectral–spatial features of the HSI cube. Denote

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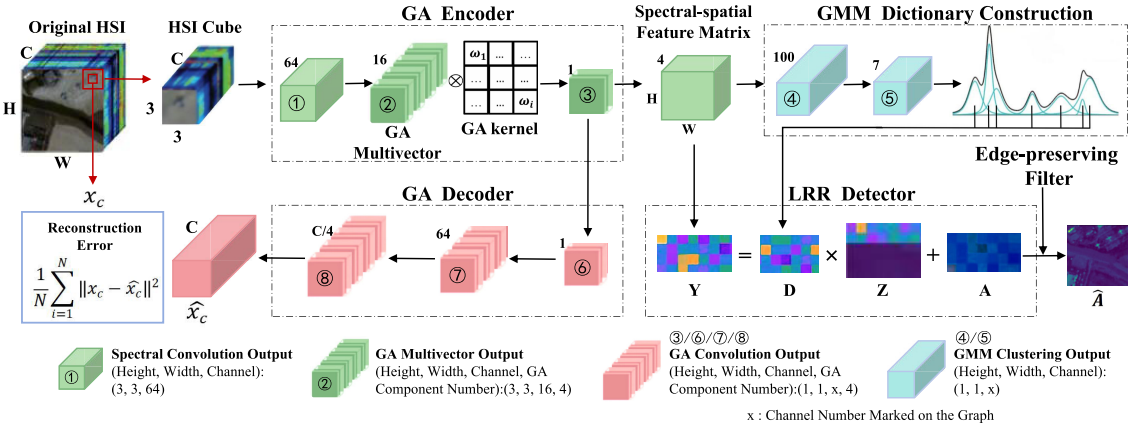


Fig. 3. Flowchart of the proposed method in the training stage.

by $\mathbf{Y} = [\mathbf{y}_i \in \mathbb{R}^s]_{i=1}^N$ the spectral–spatial features, where s is the number of output channels in the fully connected layer.

The GA encoder processes the spectral–spatial features of each pixel into a GA multivector $\widehat{y}_i$ using (9). This multivector is then fed into the GA decoder, which consists of two GA linear layers. The output of the GA linear layer, $\widehat{x}_i$, can be calculated as follows:

$$\widehat{x}_i = g\left(\sum_j^v \left(\omega_j \otimes \widehat{y}_{ij} + b_j\right)\right), \quad i = 1, \ldots, N \tag{11}$$

where v is the number of input channels and $\omega_j, b_j \in \mathbb{G}_n$. The GA form features output from the two GA linear layers are then flattened to obtain the reconstruction pixels.

Particularly, the reconstructed vector $\widehat{x}_c$ of the GA decoder represents individual pixels rather than HSI cubes. The HAD study focuses primarily on the features of the original individual background and anomaly pixels. The reconstruction error is defined as the error between the center pixel x_c of the original HSI cube and the reconstructed individual vector $\widehat{x}_c$. The reconstruction error of our GA codec can be calculated as follows:

$$\text{Reconstruction error} = \frac{1}{N}\sum_{i=1}^{N}\left\|x_c - \widehat{x}_c\right\|^2. \tag{12}$$

The network parameters of the GA codec module are detailed in Table I.

B. Dictionary Construction Module

The HSI latent feature map $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_N] \in \mathbb{R}^{s\times N}$ is obtained by extracting the spectral–spatial features of the raw HSI in the GA codec module and concatenating these features of each pixel. For the low-rank HAD inference method, building a suitable background dictionary is essential. In recent years, researchers have proposed various methods for constructing dictionaries. Among these methods, clustering dictionary construction algorithms can address the issue of accurately representing the background without prior information. In our proposed GA-HAD method, an unsupervised clustering algorithm based on the GMM [61] is used to construct a background dictionary required for inputs to an LRR model.

The GMM uses a linear combination of K single Gaussian distribution models to describe the data distribution. It is assumed that the distribution of $\mathbf{y}_i$ obeys GMM. The probability density function of $\mathbf{Y}$ is defined as

$$p(\mathbf{Y}) = \sum_{t=1}^{K}\alpha_t \mathcal{N}(\mathbf{Y}|\mu_t, \theta_t)$$
$$\text{s.t.} \sum_{t=1}^{K}\alpha_t = 1, 0 \le \alpha_t \le 1 \tag{13}$$

where α_t is the proportion of the tth Gaussian components, $\mathcal{N}(\mathbf{Y}|\mu_t, \theta_t)$ represents the tth Gaussian distribution, and μ_t and θ_t represent the mean vector and the covariance matrix of the tth Gaussian distribution, respectively.

The parameters of the GMM are trained together with the GA codec in the overall network framework using the following equations:

$$\alpha_t = \frac{\sum_{i=1}^{N}\gamma_{\text{it}}}{N}$$
$$\mu_t = \frac{\sum_{i=1}^{N}\gamma_{\text{it}}\mathbf{Y}}{\sum_{i=1}^{N}\gamma_{\text{it}}}$$
$$\theta_t = \frac{\sum_{i=1}^{N}\gamma_{\text{it}}(\mathbf{Y}-\mu_t)(\mathbf{Y}-\mu_t)^T}{\sum_{i=1}^{N}\gamma_{\text{it}}} \tag{14}$$

where γ_{it} is the probability distribution of the tth component of $\mathbf{y}_i$ and is estimated by a density estimation model in our work. The density estimation model is used to estimate the probability of which Gaussian component distribution each pixel obeys. Parameters α_t, μ_t, and θ_t can be computed by maximizing the log-likelihood function method, such as the EM algorithm.

In GMM model training, the negative log-likelihood function is used as the loss function for training

$$\text{GMM loss function} = -\frac{1}{N}\sum_{i=1}^{N}\log\sum_{t=1}^{K}\alpha_t\mathcal{N}(\mathbf{Y}|\mu_t, \theta_t). \tag{15}$$

The GMM model parameters are trained alongside those of the GA codec. As the parameters are optimized, we select the two atoms with the highest and lowest posterior

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TABLE I
NETWORK PARAMETERS OF THE GA CODEC

Network Layer	Input	Encoder1	GA Encoder2	FC Encoder3	GA Decoder1	GA Decoder2
Parameter	$3 \times 3 \times C$	$9 \times 1 \times 64$	$3 \times 3 \times 4$	4	$4 \times 1 \times 64$	$4 \times 1 \times (C/4)$

probabilities in each cluster created by the GMM. These atoms are used to construct a suitable background dictionary $\mathbf{D} = [\mathbf{y}_{d_1}, \ldots, \mathbf{y}_{d2K}] \in \mathbb{R}^{s \times 2K}$, where $\mathbf{y}_{d_1}, \ldots, \mathbf{y}_{d2K}$ are background dictionary atoms selected from $\mathbf{Y}$.

C. LRR Inference Module

After obtaining the spectral–spatial feature maps of the HSI through the GA codec and the background dictionary from the GMM, the LRR method is selected to detect anomalies from the background. LRR describes the global structure of data, and it can extract anomalies that have sparse characteristics within a background exhibiting low-rank properties. Based on LRR, the 1-D hyperspectral data (i.e., the spectral–spatial feature maps $\mathbf{Y}$ extracted by the GA codec) can be separated into background and anomaly matrices. The background part is described by the representation coefficient matrix with the background dictionary. The LRR model can be defined as follows:

$$\mathbf{Y} = \mathbf{DZ} + \mathbf{A} \tag{16}$$

where $\mathbf{Z} \in \mathbb{R}^{2K \times N}$ denotes the coefficient matrix, while $\mathbf{A} \in \mathbb{R}^{s \times N}$ denotes the anomaly matrix.

As the anomaly only occupies a small portion of the image, it can be considered sparse in $\mathbf{A}$. However, in the HSI feature map, the background pixels have a high correlation, allowing them to be linearly represented by a small number of atoms in the background dictionary, and it follows that the coefficient matrix $\mathbf{Z}$ will have sparse characteristics.

LRR aims to minimize the rank of the coefficient matrix $\mathbf{Z}$. According to matrix completion theory, the convex envelope of a matrix rank is the nuclear norm of the matrix. Therefore, the optimization problem of LRR can be transformed into a convex optimization problem

$$\min_{\mathbf{Z},\mathbf{A}} \|\mathbf{Z}\|_* + \lambda \|\mathbf{A}\|_{2,1}, \quad \text{s.t. } \mathbf{Y} = \mathbf{DZ} + \mathbf{A} \tag{17}$$

where $||\mathbf{Z}||_*$ denotes the nuclear norm of $\mathbf{Z}$ and $||\mathbf{A}||_{2,1}$ is the $l_{2,1}$ norm of $\mathbf{A}$. $\lambda > 0$ is the regularization parameter, which strikes a balance between the low-rank and sparse terms. $l_{2,1}$ norm is the result of summing the l_2 norm of each column in anomaly matrix $\mathbf{A}$

$$\|\mathbf{A}\|_{2,1} = \sum_{j=1}^{N} \sqrt{\sum\nolimits_{i=1}^{s} \alpha_{ij}^2} \tag{18}$$

where a_{ij} is the element of the ith row and jth column of matrix $\mathbf{A}$. The objective of $l_{2,1}$ norm is to minimize the number of nonzero columns, reflecting the sparseness of $\mathbf{A}$.

The optimization problem of (17) can be solved by the alternating direction method proposed by Liu et al. [62]. First, an auxiliary matrix $\mathbf{J}$ is introduced instead of the coefficient matrix $\mathbf{Z}$, and then the optimization objective is converted into the following form:

$$\begin{aligned} \min_{\mathbf{Z},\mathbf{A},\mathbf{J},Y_1,Y_2} & \|\mathbf{J}\|_* + \lambda \|\mathbf{A}\|_{2,1} + tr\left[Y_1^T(\mathbf{Y} - \mathbf{DZ} - \mathbf{A})\right] \\ & + tr\left[Y_2^T(\mathbf{Z} - \mathbf{J})\right] + \frac{\mu}{2}\left(\|\mathbf{Y} - \mathbf{DZ} - \mathbf{A}\|_F^2 + \|\mathbf{Z} - \mathbf{J}\|_F^2\right) \end{aligned} \tag{19}$$

where Y_1 and Y_2 are Lagrange multipliers, $\mu > 0$ represents the penalty parameter, $tr(\cdot)$ refers to the trace of a matrix, and $||\cdot||_F$ denotes the Frobenius norm.

Equation (19) can be solved by alternatively updating only one variable at one time while fixing the others.

1) *J Step:* Fix $\mathbf{Z}$ and $\mathbf{A}$ to update $\mathbf{J}$

$$\min_{\mathbf{J}} \frac{1}{\mu}\|\mathbf{J}\|_* + \frac{1}{2}\|\mathbf{J} - (\mathbf{Z} + Y_2/\mu)\|_F^2. \tag{20}$$

2) *Z Step:* Fix $\mathbf{J}$ and $\mathbf{A}$ to update $\mathbf{Z}$

$$\mathbf{Z} = \left(\mathbf{D}^T\mathbf{D} + \mathbf{I}\right)^{-1}\left(\mathbf{D}^T\mathbf{Y} - \mathbf{D}^T\mathbf{A} + \mathbf{J} + \left(\mathbf{D}^T Y_1 - Y_2\right)/\mu\right). \tag{21}$$

3) *A Step:* Fix $\mathbf{J}$ and $\mathbf{Z}$ to update $\mathbf{A}$

$$\min_{\mathbf{A}} \frac{\lambda}{\mu}\|\mathbf{A}\|_{2,1} + \frac{1}{2}\|\mathbf{A} - (\mathbf{Y} - \mathbf{DZ} + Y_1/\mu)\|_F^2. \tag{22}$$

4) *Y_1, Y_2 Step:* Update Y_1 and Y_2

$$\begin{aligned} Y_1 &= Y_1 + \mu(\mathbf{Y} - \mathbf{DZ} - \mathbf{A}) \\ Y_2 &= Y_2 + \mu(\mathbf{Z} - \mathbf{J}). \end{aligned} \tag{23}$$

After decomposing the HSI feature map $\mathbf{Y}$ using the optimized solution mentioned above, we obtain the anomaly matrix $\mathbf{A}$. To further refine the detection results, we utilize an edge-preserving filtering method that takes into account the spatial relationship between neighboring pixels. Specifically, we choose a bilateral filter that can remove noise while preserving edges. Denote by $\widehat{\mathbf{A}}$ the filtered anomaly matrix, the anomaly score for the ith pixel can be determined as the sum of the absolute values of all elements in column i of the anomaly matrix $\widehat{\mathbf{A}}$.

V. EVALUATION

In this section, we first introduce our evaluation setup (Section V-A) and then we present the overall performance of the GA-HAD framework (Section V-B). Furthermore, we analyze the effect of each key component in GA-HAD (Section V-C).

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TABLE II
DETAILED INFORMATION OF DIFFERENT DATASETS

Datasets	Sensor	Spatial Resolution	Image Size	Anomaly Proportion
Salinas	AVIRIS	$3.7m$	$120\times120\times204$	0.17%
Gainesville	AVIRIS	$3.5m$	$100\times100\times191$	0.52%
HYDICE	HYDICE	$1.56m$	$100\times80\times175$	0.26%
Cat Island	AVIRIS	$17.2m$	$150\times150\times188$	0.08%

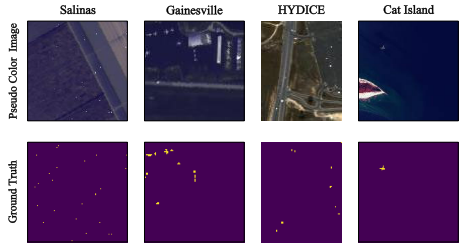


Fig. 4. Pseudo-color images and ground truth of the four datasets.

A. Experimental Setup

1) Datasets: To demonstrate the effectiveness of our GA-HAD framework, we select datasets suitable for anomaly detection, following the recommendations of [63] to avoid large visible anomalies. Specifically, we choose the datasets with either single-pixel or multipixel small anomalies: Salinas, Gainesville, HYDICE, and Cat Island.

Salinas Dataset: The first dataset is captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor. It has a spatial size of 120 × 120 pixels and consists of 204 spectral bands with a spatial resolution of 3.7 m/pixel. The dataset contains several simulated targets as anomalies, accounting for 0.17% of the total pixels.

Gainesville Dataset: The second dataset is acquired by the AVIRIS sensor and has a spatial dimension of 100 × 100 pixels with 191 spectral bands. The spatial resolution is 3.5 m/pixel. This dataset treats buildings as anomalies, which constitute 0.52% of the total pixels.

HYDICE Dataset: The third is collected by the hyperspectral digital imagery collection experiment (HYDICE) sensor and has a spatial dimension of 100 × 80 pixels with 175 spectral bands. The spatial resolution is 1.56 m/pixel. In this dataset, roofs and cars are considered anomalies, representing 0.26% of the total pixels.

Cat Island Dataset: The fourth dataset acquired by the AVIRIS sensor has a spatial size of 150 × 150 pixels and includes 188 spectral bands. It has a spatial resolution of 17.2 m/pixel. This dataset is a beach scene with anomalies representing 0.08% of the total pixels.

More detailed information about the selected datasets is provided in Table II. In Fig. 4, we present the pseudo-color images and the ground-truth anomalous pixels of different datasets.

2) Evaluation Metrics: In our experiments, we employ a combination of qualitative and quantitative evaluation metrics to assess the comprehensive performance of GA-HAD. Qualitative evaluation metrics include the color anomaly detection map, a specialized 3-D receiver operating characteristic (ROC) curve proposed in [63] specifically for HAD tasks, and box plots [64]. From the 3-D ROC curve, we can derive three distinct 2-D ROC curves: a 2-D ROC curve of (PD, PF), a 2-D ROC curve of (PD, τ), and a 2-D ROC curve of (PF, τ). For quantitative assessment, we utilize the area under the ROC curve (AUC), adopting specific AUC metrics tailored for HAD tasks as introduced in [63]. These metrics include AUC(PD, PF), AUC(PD, τ), AUC_{BDP}, and AUC_{ADBS}. Detailed explanations of evaluation metrics are as follows:

1) *Color anomaly detection map* visualizes the detection results of different HAD methods, providing an intuitive representation of anomaly responses and background suppression effects. It is the result of visualizing normalized anomaly values using a color map.
2) *3-D ROC curve* [63] is used to evaluate the detection performance by simultaneously considering the probability of detection (PD), false alarm rate (PF), and the threshold parameter τ. PD measures the ability to accurately identify true targets, while PF indicates the likelihood of misclassifying background pixels as targets. This method treats τ as an independent variable and calculates the corresponding values of PD and PF for different values of τ. By plotting these variables in a 3-D space defined by (PD, PF, τ), the 3-D ROC curve is generated. From this 3-D curve, three distinct 2-D ROC curves can be derived: one for (PD, PF), another for (PD, τ), and a third for (PF, τ).
3) *Box Plots* are commonly used to visually illustrate the central tendency and dispersion of data. In the study of HAD, box plots can be used to visually demonstrate the separability between background and anomalies. The body of a box plot consists of a rectangular box, the top and bottom of which correspond to the 75th (Q3) and 25th (Q1) percentiles of the data, respectively. The center line of the box represents the median of the data. From the top and bottom of the box are lines known as "whiskers." The whiskers typically extend to caps, which are the furthest data points within 1.5× the interquartile range (IQR $= Q3 - Q1$) from the ends of the box. Specifically, the upper whisker extends to the cap at the largest value that is less than or equal to $Q3+1.5\times\text{IQR}$, and the lower whisker extends to the cap at the smallest value that is greater than or equal to $Q1 - 1.5 \times \text{IQR}$.
4) *The AUC scores* quantify the effectiveness of the three 2-D ROC curves derived from the 3-D ROC curve. Specifically, the AUC(PD, PF) score quantifies overall detection performance by evaluating the tradeoff between PD and PF. The AUC(PD, τ) score measures the capability of the model to correctly identify anomaly pixels. The AUC(PF, τ) score indicates the performance of the model in suppressing the background and reducing false alarms. To further evaluate the overall performance in background detection, joint anomaly detection, and background suppression, additional metrics introduced in [63] are employed: AUC_{BDP} and AUC_{ADBS}. Notably, AUC(PF, τ) represents the probability of background pixels being erroneously detected as anomalies, while AUC_{BDP}, defined as 1-

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AUC(PF, τ), quantifies the probability of correctly detecting background samples. This renders AUC_{BDP} a more straightforward metric for assessing background detection performance, which is the reason why AUC_{BDP} is selected over AUC(PF, τ). Considering the impact of AUC(PF, τ) on AUC(PD, τ), the metric AUC_{ADBS} = AUC(PD, τ) - AUC(PF, τ) is used to evaluate the overall performance of joint anomaly detection and background suppression. In summary, this study employs four types of AUC scores: AUC(PD, PF), AUC(PD, τ), AUC_{BDP}, and AUC_{ADBS}, to comprehensively evaluate the performance of detection methods. The following sections offer a detailed description of these four metrics

$$\begin{cases} 0, & \leq \text{AUC(PD, PF)} \leq 1 \\ 0, & \leq \text{AUC(PD, } \tau) \leq 1 \\ 0, & \leq \text{AUC}_{\text{BDP}} = 1 - \text{AUC(PF, } \tau) \leq 1 \\ -1, & \leq \text{AUC}_{\text{ADBS}} = \text{AUC(PD, } \tau) - \text{AUC(PF, } \tau) \leq 1. \end{cases} \tag{24}$$

3) Baselines: We compare GA-HAD with 13 representative HAD methods, including one classic method (RX [5]), five representation-based methods (LRASR [25], LSMAD [11], AHMID [29], MTVLRR [31], and SuperRPCA [22]), five DL-based methods (LREN [6], RGAE [36], Auto-AD [37], GT-HAD [45], and DFAN [38]) and two tensor-based method (PTA [54] and PCA-TLRSR [55]). The technical details of these methods are provided in Section II.

Parameter Configuration: All comparison methods use default parameters from their original works, with minor adjustments made for different datasets to improve performance. Detailed parameter settings are given in Table III. In the context of these methods, the RX approach does not involve any parameter settings. For methods that incorporate representation models, the *detection epoch* for training the detection model is set to 300. For all DL methods, including LREN, RGAE, Auto-AD, GT-HAD, DFAN, and our GA-HAD approach, the *training epoch* of the DL network is also set to 300, with a learning rate lr of 0.0001. The LRASR technique adopts tradeoff parameters where λ is positioned at 0.1 and β is set to 0.1, alongside a cluster count K of 15 and dictionary atoms P extracted from each cluster of 20. For LSMAD, an upper limit for the rank of the low-rank matrix is established at 5, while the parameter k is determined to be 1. For the AHMID strategy, the tradeoff parameter λ is specified in the table, with α set to 1. Additionally, the cluster count K is set to 7, and dictionary atoms P are extracted from each cluster of 15. The MTVLRR method sets K to 7, P to 15, and the parameters λ to 0.7. For the SuperRPCA strategy, the parameter λ_a is set to 1, λ_b to 0.1, N to 2, S to 12. The LREN method defines a cluster count K of 7 and parameter λ is specified in the table. In RGAE, parameter λ is set individually for each case. The number of superpixels S is set to 200, and the dimension of the hidden layer n_hid is set to 150. For DFAN, the tradeoff parameters α, β, γ are set to 1. Additionally, the minimum number of points MinPts is set to 3, and the minimum distance between adjacent points Eps is set to 0.3. The PTA method incorporates tradeoff parameters α set to 0.001, τ set to 1, β set to 0.1, and μ set to 0.001. In PCA-TLRSR, the parameters λ and λ' are set to 0.06 and 0.05, respectively. The number of dimensions d is configured based on the specific dataset.

TABLE III
PARAMETER SETTINGS FOR COMPARISON METHODS ON FOUR DATASETS

Method	Param	Salinas	Gainesville	HYDICE	Cat Island
LRASR	K	15	15	15	15
	P	20	20	20	20
	β	0.1	0.1	0.1	0.1
	λ	0.1	0.1	0.1	0.1
LSMAD	$rank$	5	5	5	5
	k	1	1	1	1
AHMID	K	7	7	7	7
	P	15	15	15	15
	α	1	1	1	1
	λ	0.1	0.1	1	0.01
MTVLRR	λ	0.7	0.7	0.7	0.7
	K	7	7	7	7
	P	15	15	15	15
Super RPCA	λ_a	1	1	1	1
	λ_b	0.1	0.1	0.1	0.1
	N	2	2	2	2
	S	12	12	12	12
LREN	K	7	7	7	7
	λ	0.1	0.1	1	1
RGAE	λ	0.1	0.1	0.1	1
	S	200	200	200	200
	n_hid	150	150	150	150
DFAN	Eps	0.3	0.3	0.3	0.3
	$MinPts$	3	3	3	3
	α, β, γ	1	1	1	1
PTA	α	0.001	0.001	0.001	0.001
	β	0.1	0.1	0.1	0.1
	τ	1	1	1	1
	μ	0.001	0.001	0.001	0.001
PCA-TLRSR	λ	0.06	0.06	0.06	0.06
	λ'	0.05	0.05	0.05	0.05
	d	5	5	5	15
Ours(GA)	K	11	11	11	11
	λ	1	1	1	1

Our proposed GA-HAD comprises a GA codec, a dictionary builder, and an anomaly detector. Before training, the spatial dimension of the extracted HSI cubes is 3×3. The number of GMM clusters K is set to 11, and the anomaly detector is trained for 300 epochs. In terms of the hyperparameter λ in (17), we find that its optimal value varies for different datasets, and thus we conduct experiments to explore the impact of λ on the performance of GA-HAD for each dataset. As shown in Fig. 5, our experimental results indicate that the performance of our method stabilizes when λ is greater than 0.1. Therefore, to achieve optimal performance across all datasets, we set λ to 1 for each dataset. Our proposed GA-HAD is implemented with PyTorch on a laptop equipped with a 3.0-GHz Intel i7 CPU and, 64-GB memory. The network is trained on one NVIDIA GeForce RTX 3090 GPU.

B. Overall Detection Performance

Fig. 6 visualizes the detection pictures of different HAD methods, while Table IV displays the AUC scores for (PD,

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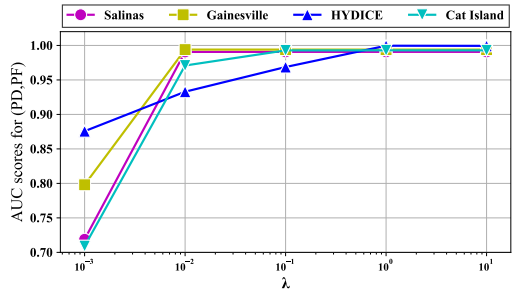


Fig. 5. AUC scores for (PD, PF) with respect to different λ settings on four datasets.

PF), (PD, τ), BDP and ADBS of compared methods. Fig. 7 shows the box plots of multiple methods. Fig. 8 provides a quantitative analysis of the separability between background and anomalies. Additionally, Figs. 9–12 present the 3-D ROC curves along with three 2-D ROC curves of compared methods for each dataset. Specifically, these 2-D ROC figures include the ROC curves of (PD, PF), (PD, τ), and (PF, τ).

Fig. 6 visualizes the detection results of different HAD methods. Our method achieves a balance between anomaly prominence and background suppression across various datasets. On the Salinas dataset, while the GT-HAD method successfully detects anomalies on the right side but overlooks those on the left, our approach closely aligns with the ground truth. For the Gainesville dataset, many methods only identify anomalies on the right side of the image, whereas our method excels at detecting anomalies, particularly on the left. Although there is some room for improvement in background suppression, our Gainesville result is effective. In the HYDICE dataset, MTVLRR and our method perform comparably well, achieving balanced detection of anomalies and background. Regarding the Cat Island dataset, GT-HAD, DFAN, and our method yield the best overall performance, effectively highlighting anomaly pixels while also achieving good background suppression.

Fig. 7 shows box plots for multiple methods on four datasets, where red boxes indicate anomalous pixels and purple boxes indicate background pixels. The height of the purple box reflects background suppression, and the distance between the red and purple boxes indicates anomaly–background separability. Fig. 8 quantifies this separability by plotting the delta between the smallest anomaly cap and the largest background cap for each method, with different colored lines representing each dataset for clear comparisons. As can be seen from Fig. 7, it is clear that our method achieves superior background suppression in the Salinas, HYDICE, and Cat Island datasets, as indicated by the lower height of the purple boxes representing the background detection outputs. In the Gainesville dataset, our method shows high anomaly detection performance. Furthermore, Fig. 8 highlights the outstanding separation performance of our method between anomalies and background, consistently ranking it among the top methods. In particular, in the HYDICE dataset, our method achieves the best separation performance. Regarding the Gainesville dataset, our method is second only to MTVLRR in terms of separation performance. Similarly, for the Cat Island dataset, our method is second only to DFAN. For the Salinas dataset, while other methods show closely clustered results, our method secures an above-average position among them.

Figs. 9–12 show the 3-D ROC curves along with three corresponding 2-D ROC curves, providing an intuitive and comprehensive evaluation of the effectiveness of different methods. The closer the 2-D ROC (PD, PF) curve is to the upper left corner, the better the overall performance. The closer the 2-D ROC (PD, τ) curve is to the top right corner, the better the method performs in detecting anomalies. Conversely, the closer the 2-D ROC (PF, τ) curve is to the lower left corner, the better the background suppression is. Our method is generally positioned toward the upper left corner in 2-D ROC (PD, PF) curves and closer to the top right corner in 2-D ROC (PD, τ) curves, indicating that our approach excels in overall performance and the detection of anomalous pixels. While the proposed method performs well on most metrics, the 2-D ROC (PF, τ) curves do not approach the bottom-left corner as closely as some comparison methods. This arises from the design of the GA convolutional layer, which focuses on preserving spectral structure and integrating spatial–spectral features. Although the use of GA enhances anomaly visibility in complex scenes, it may also render the model more susceptible to subtle spectral variations, leading to a moderate increase in PF when background regions exhibit spectral similarity to anomalies. The design reflects a tradeoff favoring detection performance over strict false alarm control. Future work may explore adaptive thresholding or regularization techniques to better balance these aspects.

Table IV presents a more quantitative assessment of the comprehensive performance of different methods, as measured by AUC(PD, PF), AUC(PD, τ), AUC_{BDP} and AUC_{ADBS} scores. Among these metrics, the best-performing methods are indicated in bold, and the second-best methods are underlined. The AUC(PD, PF) metric is used to evaluate the overall performance of a detector in discriminating between anomalies and background. Generally, GA-HAD consistently outperforms all baselines across different datasets in terms of AUC(PD, PF), achieving 99.08%, 99.40%, 99.96%, and 99.29% in the Salinas, Gainesville, HYDICE, and Cat Island datasets, respectively. On average, our proposed method improves performance by 7.14%, 4.43%, 3.71%, 5.70%, 2.07%, 9.50%, 6.20%, 12.48%, 4.82%, 8.75%, 0.88%, 15.23%, and 1.54% compared to the RX, LRASR, LSMAD, AHMID, MTVLRR, SuperRPCA, LREN, RGAE, Auto-AD, GT-HAD, DFAN, PTA, and PCA-TLRSR methods, respectively. In addition to overall performance, effective background suppression and anomaly detection are two critical metrics for evaluating the performance of a detector in HAD tasks. For background detection, the AUC_{BDP} metric is used to assess the effectiveness of background detection performance. Regarding anomaly detection, AUC(PD, τ) indicates the effectiveness in highlighting anomalous pixels. AUC_{ADBS}, on the other hand, evaluates the combined performance of anomaly detection and background suppression, taking into account the influence of AUC(PF, τ) on AUC(PD, τ). As shown in Table IV, our method achieves the best performance in the AUC(PD, PF) metric and is in the top three of all fourteen methods for both AUC(PD, τ) and AUC_{ADBS}. However, there is a slight deficiency in the AUC_{BDP} metric, which

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TABLE IV
AUC SCORES FOR DIFFERENT COMPARED METHODS ON FOUR DATASETS

Methods	Salinas				Gainesville			
	AUC(PD, PF)↑	AUC(PD, τ)↑	AUC_{BDP}↑	AUC_{ADBS}↑	AUC(PD, PF)↑	AUC(PD, τ)↑	AUC_{BDP}↑	AUC_{ADBS}↑
RX [5]	0.8189	0.2042	0.9686	0.1728	0.9496	0.0957	0.9650	0.0607
LRASR [25]	0.9463	0.4852	0.9555	0.4407	0.9340	0.3777	0.8657	0.2434
LSMAD [11]	0.9441	0.3544	0.9858	0.3402	0.9636	0.1498	0.9690	0.1188
AHMID [29]	0.9296	0.3628	0.9883	0.3511	0.9535	0.2213	0.9649	0.1862
MTVLRR [31]	0.9215	0.3960	0.9612	0.3572	0.9887	0.4614	0.9523	0.4137
SuperRPCA [22]	0.9727	**0.5996**	0.9926	**0.5922**	0.9620	0.2777	0.9765	0.2542
LREN [6]	0.9085	0.3884	0.9422	0.3306	0.9250	0.2375	0.9329	0.1704
RGAE [36]	0.9285	0.3528	0.9825	0.3353	0.8232	0.1030	0.9612	0.0642
Auto-AD [37]	0.9552	0.3968	**0.9933**	0.3901	0.8840	0.1518	0.9899	0.1417
GT-HAD [45]	0.9452	0.4040	0.9878	0.3918	0.8154	0.0738	**0.9932**	0.0670
DFAN [38]	0.9887	0.5096	0.9276	0.4372	0.9681	0.4058	0.8944	0.3002
PTA [54]	0.8653	0.5164	0.8893	0.4057	0.7879	0.3092	0.8189	0.1281
PCA-TLRSR [55]	0.9871	0.5612	0.9662	0.5274	0.9882	0.4377	0.9419	0.3796
Ours(GA)	**0.9908**	0.5320	0.9735	0.5055	**0.9940**	**0.7901**	0.8144	**0.6045**

Methods	HYDICE				Cat Island			
	AUC(PD,PF)↑	AUC(PD,τ)↑	AUC_{BDP}↑	AUC_{ADBS}↑	AUC(PD,PF)↑	AUC(PD,τ)↑	AUC_{BDP}↑	AUC_{ADBS}↑
RX [5]	0.9761	0.2166	0.9620	0.1786	0.9675	0.2432	0.9925	0.2357
LRASR [25]	0.9986	0.4837	0.9306	0.4143	0.9371	0.2332	0.9557	0.1889
LSMAD [11]	0.9888	0.1745	0.9834	0.1579	0.9386	0.1121	0.9941	0.1062
AHMID [29]	0.9587	0.1461	0.9795	0.1256	0.9209	0.1195	0.9909	0.1104
MTVLRR [31]	0.9991	0.3968	**0.9925**	0.3893	0.9872	0.2116	0.9903	0.2019
SuperRPCA [22]	0.7317	0.1629	0.9355	0.0984	0.9659	0.2326	0.9900	0.2226
LREN [6]	0.9590	0.3747	0.8986	0.2733	0.9525	0.3505	0.9601	0.3106
RGAE [36]	0.8402	0.2150	0.9074	0.1224	0.9440	0.1055	0.9818	0.0873
Auto-AD [37]	0.9854	0.2576	0.9812	0.2388	0.9697	0.1568	**0.9943**	0.1511
GT-HAD [45]	0.9039	0.1545	0.9584	0.1129	0.9929	0.4100	0.9894	0.3994
DFAN [38]	0.9976	**0.6700**	0.8706	**0.5406**	0.9884	**0.7134**	0.9520	**0.6654**
PTA [54]	0.8598	0.4239	0.8197	0.2436	0.9385	0.2181	0.9727	0.1908
PCA-TLRSR [55]	0.9901	0.6095	0.9131	0.5226	0.9515	0.3068	0.9804	0.2872
Ours(GA)	**0.9996**	0.5474	0.9632	0.5106	**0.9929**	0.3942	0.9752	0.3694

may be due to the complexity of the background variations, indicating potential space for improvement in dictionary construction.

Compared with the DL method, LREN, our method further optimizes the feature extraction method by introducing GA convolutional layers to extract spectral–spatial information. This innovation not only enhances the model's ability to capture anomalous features but also significantly improves the overall performance of the detector. The RGAE method is more demanding in terms of datasets and network parameters, resulting in unstable performance on these datasets. The DL-based Auto-AD method has excellent background detection capabilities. However, the overall performance is slightly degraded due to its less-than-ideal anomaly detection performance. In contrast to another DL method, GT-HAD, our method overcomes the defect of GT-HAD missing some important anomaly information due to the suppression of background. Specifically, our method has achieved an average 104.92% improvement in AUC_{ADBS} score, contributing to a 8.75% average increase in AUC(PD, PF), enhancing overall performance. When compared to DFAN, our method shows a better overall AUC_{BDP} with a 2.24% improvement, indicating superior background suppression capabilities. In addition, with similar AUC(PD, τ) and AUC_{ADBS}, we achieved better overall performance across all scenario datasets, with AUC(PD, PF) improved by 0.88%.

TABLE V
SETTINGS OF ABLATION METHODS

Model	Feature Extraction	Dictionary Construction	LRR	Filter	Train together
(A)	✘	✔	✔	✔	✔
(B)	MLP	✔	✔	✔	✔
(C)	CNN	✔	✔	✔	✔
(D)	✔	K-means	✔	✔	✔
(E)	✔	✘	✘	✔	✔
(F)	✔	✔	✔	✘	✔
(G)	✔	✔	✔	✔	✘

In summary, despite sacrificing some background purity, GA-HAD achieves the best overall performance among the 14 comparative methods.

C. Ablation Study

In this section, we conduct a series of ablation experiments to analyze the individual impact of each key component in GA-HAD. The experimental setup is detailed in Table V, and the results are presented in Table VI.

1) Feature Extraction Through the GA Codec: To analyze the individual impact of the GA feature extraction module on HAD performance, we replace it with three other methods in additional experiments.

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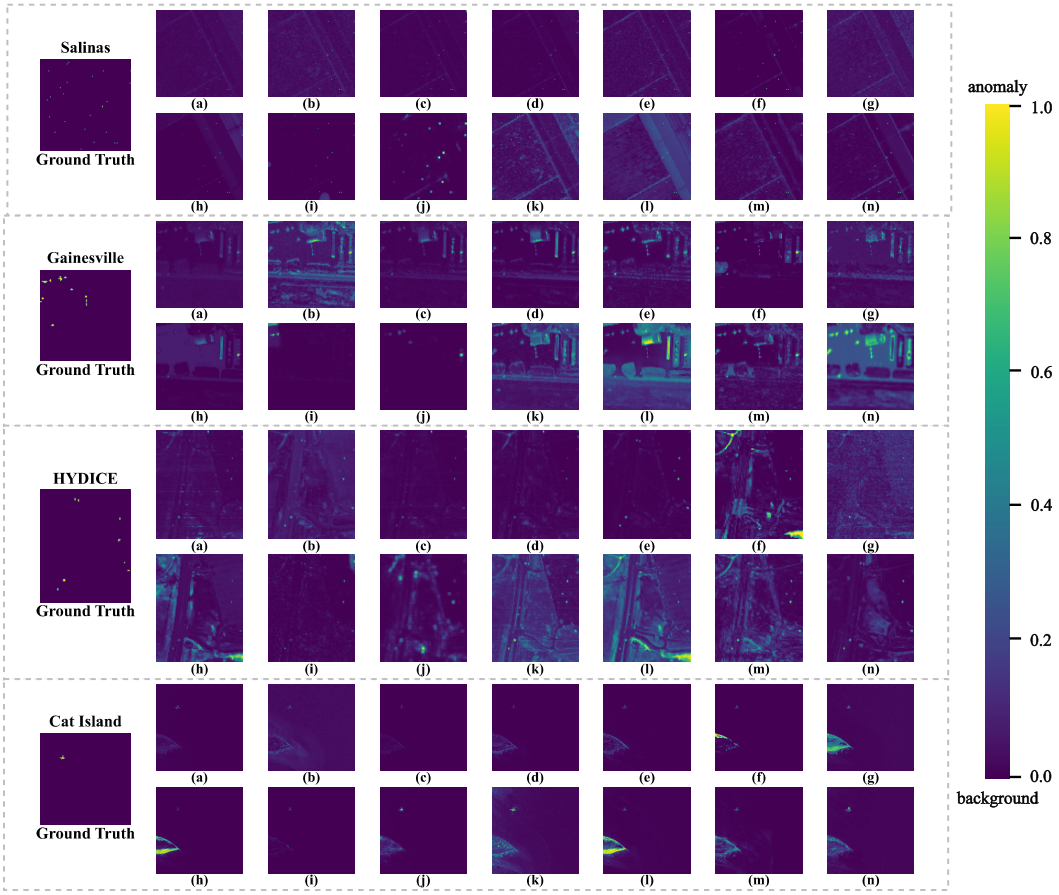


Fig. 6. Ground-truth maps and detection maps of the comparison methods on four datasets. (a) RX. (b) LRASR. (c) LSMAD. (d) AHMID. (e) MTVLRR. (f) SuperRPCA. (g) LREN. (h) RGAE. (i) Auto-AD. (j) GT-HAD. (k) DFAN. (l) PTA. (m) PCA-TLRSR. (n) Ours(GA).

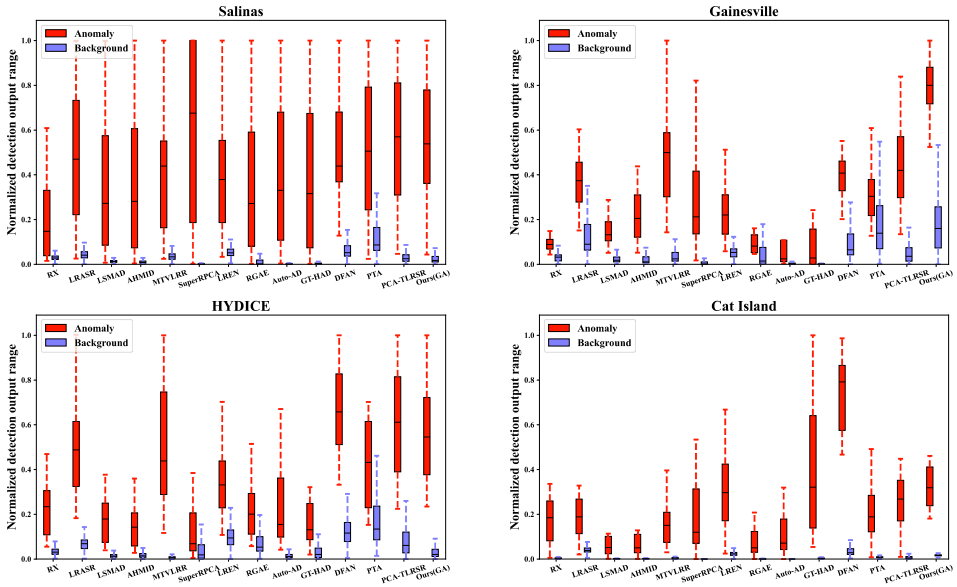


Fig. 7. Box plots of the comparison methods on four datasets.

1) *Experiment (A)* directly applies the LRR method in the original HSI space without the need for feature extraction.
2) *Experiment (B)* replaces the GA module with multilayer perception (MLP) for extracting the spectral feature of individual pixels.
3) *Experiment (C)* substitutes the GA module with a real-valued CNN to extract the spectral–spatial feature.

To provide a clear and fair illustration of the benefits brought by the GA feature extraction network, Experiments (A)–(C) employ distinct feature extraction techniques while maintaining the subsequent processing and model

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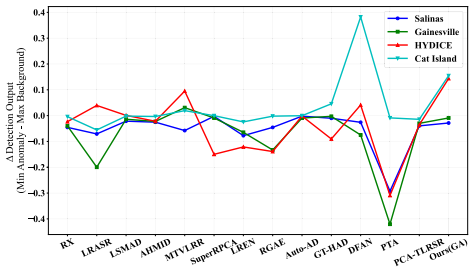


Fig. 8. Detection output Δ (min anomaly − max background) in box plots.

TABLE VI
AUC SCORES FOR (PD, PF) FOR DIFFERENT ABLATION METHODS ON FOUR DATASETS

Model	Salinas	Gainesville	HYDICE	Cat Island
(A)	0.7833	0.4239	0.5151	0.8619
(B)	0.3880	0.9372	0.8517	0.6996
(C)	0.9703	0.4570	0.8428	0.8173
(D)	0.8060	0.8250	0.9890	0.9458
(E)	0.9590	0.9667	0.9745	0.9483
(F)	0.9860	0.9921	0.9921	0.9881
(G)	0.9856	0.8131	0.9844	0.9344
Ours(GA)	**0.9907**	**0.9940**	**0.9996**	**0.9929**

structures unchanged. To ensure a fair comparison, the CNN in Experiment (C) adopted the same 3 × 3 kernel size as the GA convolutional layer used in the GA-HAD method.

Fig. 13 compares the AUC performance of three experiments with GA-HAD, where the solid black line represents our proposed method, GA-HAD, with the numerical value above it, indicating the AUC score. The values beneath each histogram indicate the specific decrease in AUC scores for the corresponding ablation study methods compared to GA-HAD. Fig. 13 shows that the use of different types of feature extraction methods imposes stricter requirements on the datasets. Apart from the real-valued CNN method, which performs well on the Salinas dataset, the other experiments do not show good effects. The poor performance of the featureless extraction method can be attributed to the sensitivity of the selected datasets to the parameter settings. Specifically, the parameters of the subsequent GMM and LRR modules have a significant impact on the effectiveness of the method. In contrast, the poor performance of using MLP and CNN can be attributed to the demanding requirements for parameter selection and training in trainable network models. Compared to MLP, CNN further decreases the differences between background pixels and abnormal pixels due to the introduction of spatial information, which is the primary limitation of real-valued convolution. The GA algorithm encoder makes full use of the spatial information of the HSI and can also fully preserve the inherent structure between spectral bands, which improves the limitation of real-valued convolution and thus improves the detection results. In the datasets that exhibit the most favorable outcomes in the featureless extraction method, MLP, and CNN experiments, our GA convolutional feature extraction method also improves AUC scores by 15.20%, 6.06%, and 2.10%, respectively.

2) Dictionary Construction and LRR Inference: To verify the necessity of the dictionary construction module and the LRR inference, we design the following experiments.

1) *Experiment (D)* selects K-means method for dictionary construction module.
2) *Experiment (E)* removes the LRR module and determines the exception directly based on the reconstruction error of the GA codec.

In Experiment (D), GMM surpassed K-means, achieving a notable 11.54% average boost in target detection accuracy. This improvement is due to GMM's adaptability and parameter optimization for background dictionaries. Its strengths enhance the handling of complex backgrounds, leading to more precise dictionaries and improved target detection. Experiment (E) demonstrates that the LRR method improves model detection accuracy by an average of 3.34%. Compared to directly assessing anomalies through reconstruction errors, the LRR method can more accurately characterize the low-rank properties of the background, enabling a more precise identification of anomalies.

3) Other Operations: Furthermore, we conduct additional experiments to discover the impact of another two operations.

1) *Experiment (F)* removes edge-preserving filtering. Finally, we experimented.
2) *Experiment (G)* separates the training of feature extraction and dictionary construction.

Experimental results indicate that edge-preserving filtering further exploits the spatial information inherent in HSIs, resulting in an average 0.48% improvement in detection. From the results of Experiment (G), it can be observed that separating the training process of feature extraction and the dictionary construction module results in a significant performance degradation, averaging about 6.99%. This is because the separation of training for feature extraction and dictionary construction presents a big challenge for the system. The optimal solution for feature extraction may not meet the needs of dictionary construction, potentially reducing system performance. Specifically, this separation causes each module to seek its own local optimum, ignoring their interrelationships.

D. Time Efficiency Analysis

In this section, we conducted a comprehensive test of the running times of various methods. Specifically, for methods that involve a training process, we separately recorded their training and detection times. Scatter plots, as shown in Fig. 14, are provided to visualize the tradeoff between model accuracy and time cost. The x-axis indicates the training time, the y-axis shows the AUC(PD, PF) score, and the size of each point reflects the detection time, with exact values listed in the legend. Since the RX method does not have a training process, we designate its detection time as its training time.

Analyzing the data in Fig. 14, it is clear that although our method outperforms the baseline methods in terms of experimental detection results, the inclusion of the GA module has appropriately increased the system overhead. We observe that our method outperforms most of the baseline methods in terms of detection time and reaches the median in terms of inference and training times. The improved detection accuracy

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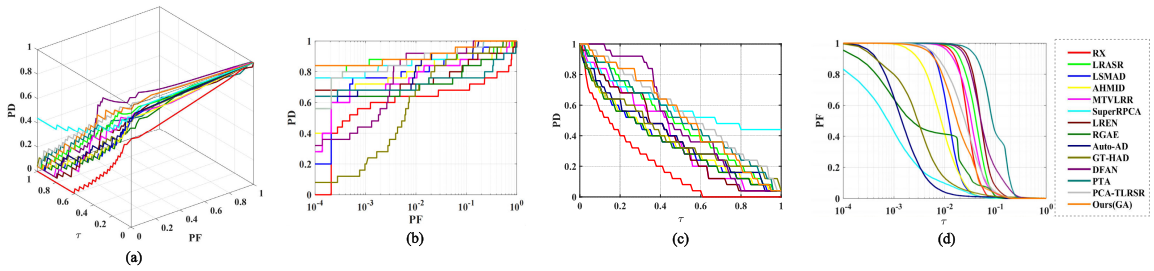


Fig. 9. 3-D ROC curve and three 2-D ROC curves for different compared methods on the Salinas dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

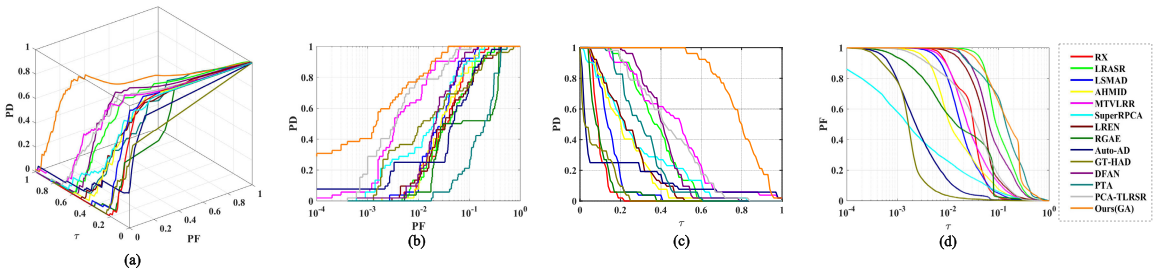


Fig. 10. 3-D ROC curve and three 2-D ROC curves for different compared methods on the Gainesville dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

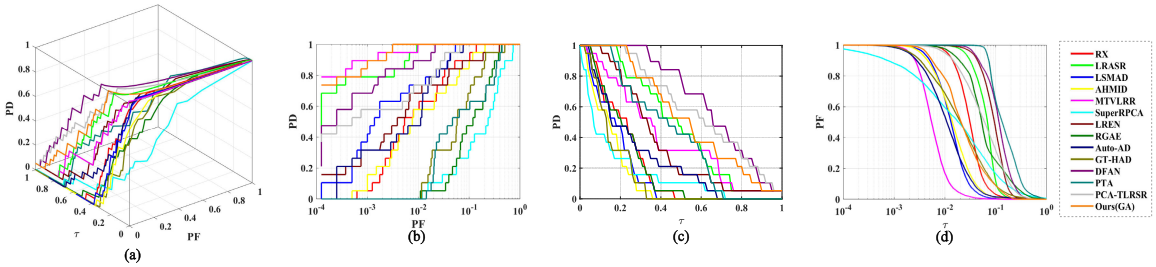


Fig. 11. 3-D ROC curve and three 2-D ROC curves for different compared methods on the HYDICE dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

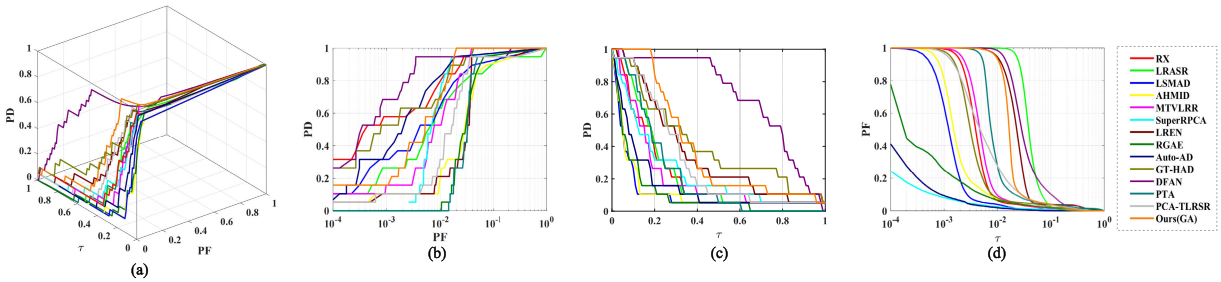


Fig. 12. 3-D ROC curve and three 2-D ROC curves for different compared methods on the Cat Island dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

from GA motivates us to further optimize system time for better efficiency and performance balance.

E. Model Complexity

In this section, because DL and non-DL methods are implemented on different software platforms, an entirely fair comparison of computational complexity is challenging. Therefore, to provide a more equitable analysis of the computational efficiency of GA-HAD, we only compare it with six DL-based HAD methods in terms of model complexity and parameter count.

The results are illustrated in Fig. 15, where each point represents a different method, and the size of the points along with the annotated values indicates the number of model parameters in millions (Params). The x-axis shows the computational complexity measured in Giga floating point operations (GFLOPs), while the y-axis represents the AUC(PD, PF) scores. As can be observed in Fig. 15, our proposed GA-HAD achieves the best detection performance with an AUC score of 0.9908, while requiring the lowest computational complexity (0.087 GFLOPs) and the fewest model parameters (0.051M) among all compared DL-based methods. These

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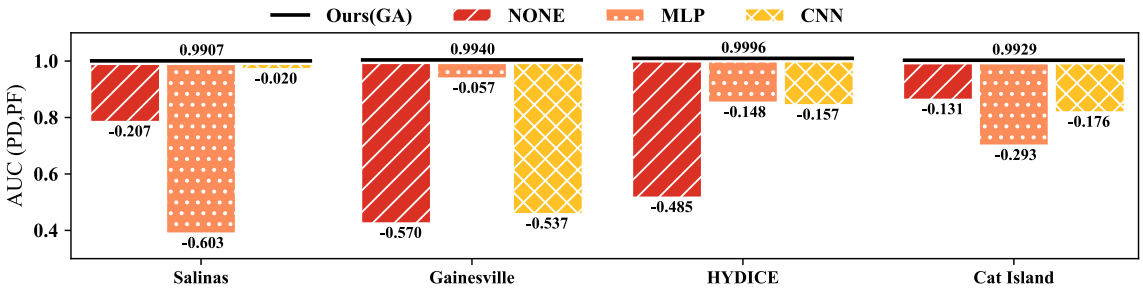


Fig. 13. Comparison of AUC scores of different feature extraction methods.

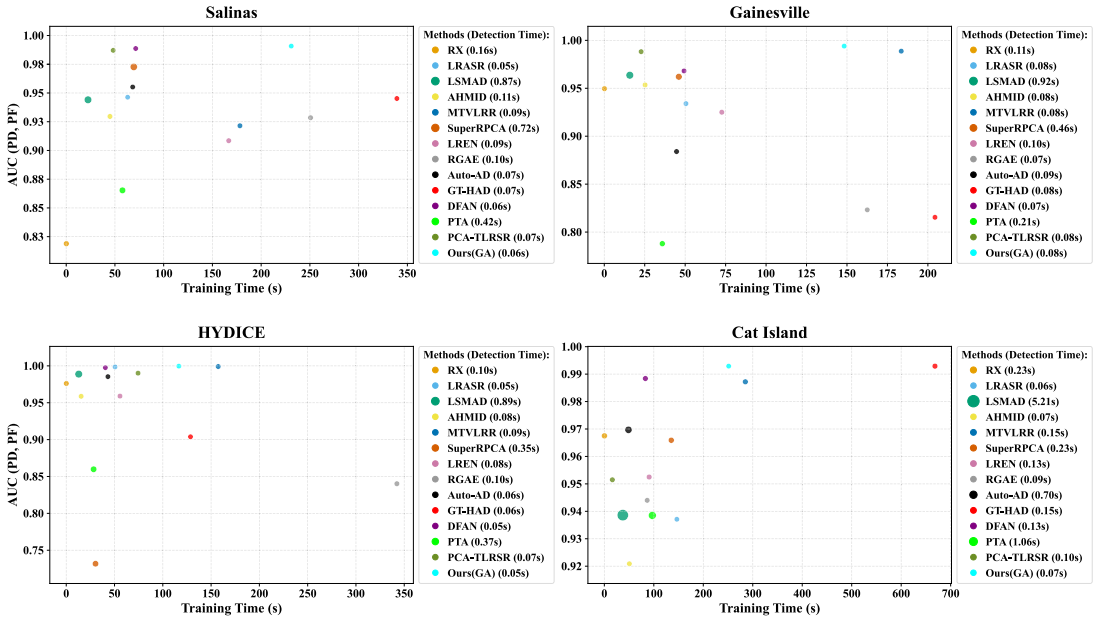


Fig. 14. Running time of the comparative methods on four datasets.

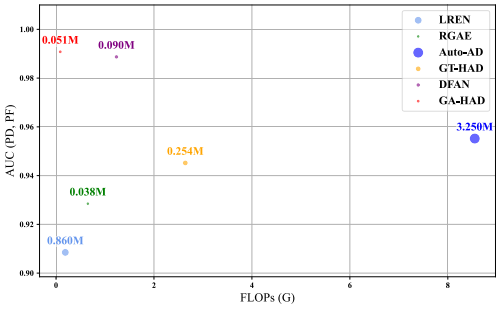


Fig. 15. Model complexities of six DL-based methods.

results highlight the effectiveness of GA-HAD in achieving high detection accuracy with significantly reduced model complexity and computational overhead.

VI. Conclusion

In this article, we propose a trainable DL network named GA-HAD, specifically designed for HAD tasks. To enhance the distinction between background and anomalies, we employ the GA-CNN to extract the spectral–spatial features of HSIs, preserving their inherent spectral structure information during the training process of the feature extraction network. To further improve the robustness of the model, we design a proprietary reconstruction error for training the GA feature extraction network and integrate it with the LRR method. Additionally, to make better use of spatial information, we apply edge-preserving filtering to refine the initial detection results, significantly enhancing the detection performance. Experimental results show that our method outperforms several state-of-the-art HAD methods on four datasets in different scenarios. Ablation experiments further confirm the rationality of the design, which effectively improves the discrimination between background and anomalies.

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